

# Thermal comfort driven space heating control via hierarchical state machine strategy interacting with multiphysics simulation

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## ABSTRACT

Improving energy efficiency of space heating systems is as important as adopting emerging clean energy technologies for decarbonising heating. While exploring strategies for reducing energy consumption, the space occupants' expectations to thermal comfort should not be sacrificed. Conventional space heating control systems are often developed under the hypothesis of uniform temperature distribution in a bounded space and the heating system controller is primarily operated based on the information from the temperature sensor at a fixed position. That is, the measured temperature does not represent the temperature surrounding the occupant's location. This paper explores a new control method; the heating system control is led by the occupant's thermal experience, expressed by the temperature surrounding the occupant. A new hierarchical state machine control strategy interacting with a Multiphysics simulation model to dynamically estimate the temperature at the point of the occupant is developed. The accuracy of the Multiphysics simulation model is verified based on the corresponding experimental test room data. A co-simulation platform is built to allow COMSOL and SIMULINK to communicate in real-time for control strategy realisation. The experimental results show that the maximum and average mean square error between the measured and simulated temperature under different operating modes is less than 0.21°C and 0.06°C, respectively. The simulation results indicate that energy consumption could be potentially reduced around 20% with an improved thermal comfort of the occupant by 30%, compared with the conventional fixed sensor control method.

## 1. Introduction

In the past ten years, CO<sub>2</sub> emissions from power sectors in the UK has been greatly reduced because of the rapid increase in power generation from renewable energy sources [1]. However, the emissions from heating are almost unchanged [2]. It is impossible to achieve the Net Zero goal without decarbonising the heating as it contributes nearly half of the total energy consumption, mainly using gas, in the UK, in which 57 % is used to meet domestic space heating and hot water demands [3,4]. Electric heating using power from renewable energy can promote heating decarbonization and proper electric heating management may support flexible grid operation to allow increased integration of renewable energy [5]. So, electrifying heating, including direct electrical heating, is considered as one of the feasible ways forward [6–8]. Currently, most of the electric heating control systems are developed based on model-free approaches, which is based on the ambient temperature measured by the temperature sensors at the fixed locations without considering the influences of the thermodynamics of the

building space and the thermal experience of the occupants. Achieving more efficient space heating control is highly appreciated in the UK, and worldwide [9]. Essentially, the space heating control system is expected to provide better scheduling to avoid wasting heat and potentially zoning in the room to allow varying space conditions [10].

To include the thermodynamic characteristics into temperature control, thermodynamic modelling is essentially required [11,12]. Thermodynamic models can be analytic with the fundamental physical mathematical equations [13], or no-analytic data driven models [14] or the mixed physical analytic equation descriptions with data driven model blocks [15,16]. In [17], a dynamic energy performance model is built with considering realistically measured physical parameters to analyse and evaluate the retrofits in buildings based on the derivation of physical equations and the assignment of values to the parameters in the model using prior knowledge. Broujeny et al. [18] reported modelling heating dynamics of buildings by a data-driven machine learning approach and then applied this model in space heating control system. The hybrid model first uses physical knowledge to build a mathematical

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model and then uses artificial Neural Networks to mine the relationships between a large amount of actual residential thermal data to obtain the values of the parameters in the model. Hu et al. [19] developed a self-learning room thermal model that uses indoor air temperature and outdoor air temperature to predict and scale model parameters. The models can provide more detailed information of the space temperature distributions and thermal flow dynamics, which can be used for control decisions.

It is popular in space heating control systems using thermostatic control [20], adaptive thermostat control [21], PID control [22], and gain control [22]. The more advanced control systems have been extensively studied in recent years, such as model predictive control [23,24], distributed model predictive control [25], fuzzy logic control [18,26]. However, most reported space heating control systems are developed based on the assumption that temperature distribution within a bounded space is uniform and the control strategy is often derived from the single sensor measurement. To address this issue, this paper aims to investigate a new heating control method by using the temperature information at the occupant location, which may improve the occupant experience of thermal comfort while using less energy.

The main contributions of the paper present in i) explored a method of building a 3D Multiphysics simulation model suitable for dynamic prediction of the temperature around the occupant position which is verified using the test data from an experimental test room; ii) established a co-simulation platform enabling the simultaneous data communication between the Multiphysics model and the control strategy in real-time; iii) development of a hierarchical state machine control system, which demonstrates around 20 % energy efficiency improvement and the occupant thermal comfort enhancement.

The remaining content of the paper is organized in the following order: The Multiphysics simulation model for space thermodynamic analysis is described in Section 2 and Section 3 presents the hierarchical state machine control strategy. The experimental system setup is illustrated in Section 4. In Section 5, the experiment test, simulation study and results analysis are given. The concluding remarks are summarized in section 6.

## 2. Multiphysics simulation model for space thermodynamic analysis

Multiphysics simulation considers the interaction process of multiple physical fields in a system with the purpose of simulating and analysing the system's behaviour more comprehensively. COMSOL Multiphysics is a software with finite element analyser and solvers built in, which is suitable for analysing engineering systems with coupled Multiphysics. COMSOL Multiphysics software is chosen in the paper for a bounded

space thermodynamic modelling to gain the space temperature distribution while considering the influence of multi-physical fields such as heat transfer in solids and fluids and fluid flow.

### 2.1. Multiphysics simulation system setup

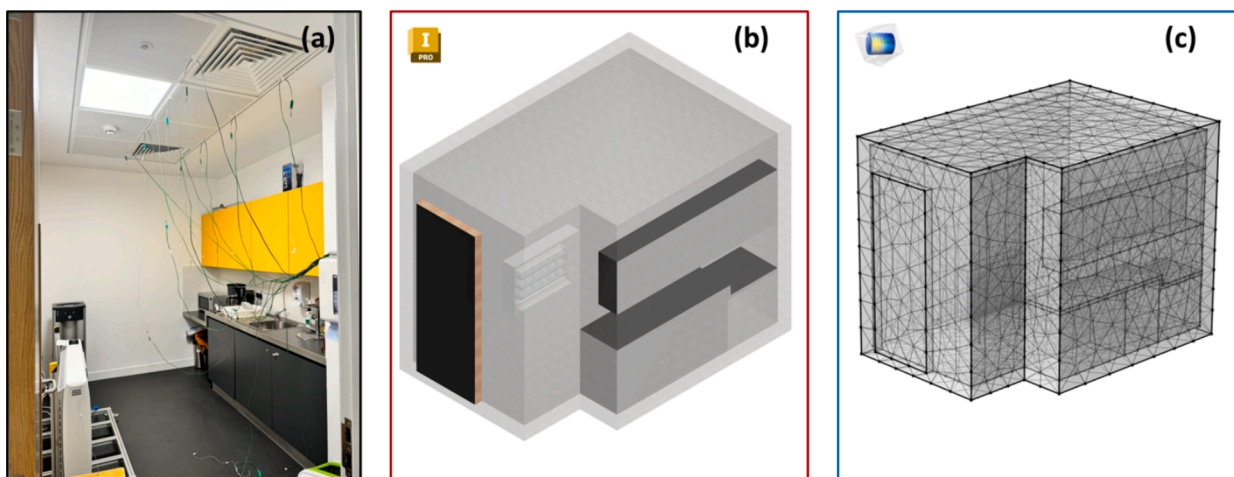
In this study, a kitchen room in the authors' research laboratory is chosen as a test space as shown in Fig. 1(a), which could represent general type of rooms in a building. The dimensions of the room are listed in Table 1. The room is assumed to be exposed to the environment with an invariable ambient temperature. Some objects inside the room are not captured in the modelling if they have little influences on simulation results. An electric storage heater is installed inside the room as a heating source. The heater contains three electric heating modules with magnetite as a storage medium and uses a fan to extract the stored heat from the bottom of the heater into the space. This storage electric heater is well insulated, so during operation, the surface temperature is just slightly different from the ambient temperature and primarily uses forced convection method to heat the space. An experiment was conducted to compare the average surface temperature of the electric heater, the ambient temperature, and the temperature of the hot air released from the bottom of the heater during its operation. Based on the test, even the heater operated at its maximum power of 3 kW, its surface temperature was only 3.1°C higher than the ambient temperature, whereas the hot air released from the heater was 28.7°C higher. With the support of these test results, the effect of radiation from the heater was not considered in the simulation to simplify the model.

A 3D geometry model of the kitchen room with enclosed furniture and appliances was built using INVENTOR software, as shown in Fig. 1 (b). INVENTOR is a 3D computer-aided design software developed by

**Table 1**

Settings and parameters used in the Multiphysics simulation model.

| Parameter [Unit]                  | Value                            |
|-----------------------------------|----------------------------------|
| Room length [m]                   | 3.5                              |
| Room width [m]                    | 2.4                              |
| Room height [m]                   | 2.65                             |
| Wall thickness [m]                | 0.11                             |
| Initial room temperature [°C]     | $T_{ini}$                        |
| Ambient temperature [°C]          | $T_{am}$                         |
| Power of the heating modules [kW] | $P_{HM1}, P_{HM2}, P_{HM3}$      |
| Air velocity from the fan [m/s]   | $V$                              |
| Relative tolerance [-]            | $10^{-3}$                        |
| Meshing technique [-]             | Physics-controlled               |
| Time stepping technique [-]       | Backward differentiation formula |
| Simulation timestep [s]           | 60                               |



**Fig. 1.** Space for simulation modelling: (a) a picture of the kitchen room; (b) 3D kitchen geometry model; (c) Multiphysics simulation model mesh.

Autodesk that provides a powerful tool to catch a 3D space. The data of the 3D space model implemented in INVENTOR is then imported into COMSOL Multiphysics through Livelink interface software package. In COMSOL Multiphysics, the material properties and boundary conditions are assigned to the 3D model. The built-in meshing tool in COMSOL is used with a physics-controlled setting for the entire geometry as shown in Fig. 1(c). The input parameters of the Multiphysics simulation model are defined as the working power of the three heating modules and air velocity from the fan of the storage electric heater. The simulation model parameter settings are listed in Table 1.

### 2.1.1. Governing mathematical equations

In COMSOL, the solution is obtained by coupling the heat transfer in solids and fluids and laminar flow physics. In this modelling and simulation study, the following assumptions are made:

- Fluid flow is assumed to be incompressible.
- Boussinesq approximation is used.
- All materials are considered homogeneous and isotropic.
- Viscous dissipation term in the energy equation is considered negligible.
- Effect of radiation heat transfer is negligible.
- The gravity is included.

Mass, momentum, and energy conservation laws are applied to the bounded physical space model to obtain the temperature distribution. For the transient problem formulation, the dependent variables of interest are temperature ( $T = f(x, y, z, t)$ ), air velocity ( $\mathbf{U} = \{u(x, y, z, t), v(x, y, z, t), w(x, y, z, t)\}$ ) and pressure, where  $x$ ,  $y$  and  $z$  represent the coordinate position in a three dimension space. The governing equations for the conservation of mass, momentum, and energy are described in Eq. (1)–(3).

The continuity equation is used to describe the conservation of mass and can be written as follows:

$$\nabla \bullet (\rho \mathbf{U}) = 0 \quad (1)$$

where,  $\rho$  is the density ( $\text{kg}/\text{m}^3$ ), which is influenced by temperature;  $\mathbf{U}$  is the air velocity vector ( $\text{m}/\text{s}$ ).

The Navier-Stokes momentum equation describes motion in a fluid and is represented in Eq. (2) for the three-dimensional case.

$$\frac{\partial(\rho \mathbf{U})}{\partial t} + \rho(\mathbf{U} \bullet \nabla) \mathbf{U} = \nabla \bullet [-p\mathbf{I} + \mu(\nabla \mathbf{U} + (\nabla \mathbf{U})')] + \rho \mathbf{g} \quad (2)$$

where,  $p$  is air pressure ( $\text{Pa}$ ).  $\mathbf{I}$  is identity matrix;  $\mu$  is the air dynamic viscosity ( $\text{Pa} \bullet \text{s}$ );  $\mathbf{g}$  is the gravity vector ( $\text{m}/\text{s}^2$ ).

For heat transfer problems, the energy balance equation can be expressed in the following form:

$$\frac{\partial(\rho C_p T)}{\partial t} + \rho C_p \mathbf{U} \bullet \nabla T = \nabla \bullet (k \nabla T) + Q \quad (3)$$

where,  $C_p$  is the specific heat at constant pressure ( $\text{J}/(\text{kg} \bullet \text{K})$ ), which is influenced by temperature;  $k$  is the thermal conductivity ( $\text{W}/(\text{m} \bullet \text{K})$ );  $Q$  is the applied external heat source ( $\text{W}/\text{m}^3$ ).

### 2.1.2. Initial and boundary conditions

The initial condition specifies the temperature field and velocity vector distributions at  $t = 0$ , and its general form is shown as follows:

$$\begin{aligned} T(x, y, z, t = 0) &= f_0(x, y, z) \\ \mathbf{U}(x, y, z, t = 0) &= \{0, 0, 0\} \end{aligned} \quad (4)$$

where,  $f_0(x, y, z)$  is the initial temperature field.

Heat flux through the walls, the roof, the floor, and the door has been modelled according to the heat transfer coefficients specified in the

European national building regulations [27]. The space outside the walls and door has been considered external. The convective heat transfer coefficients are shown in Table 2 for the exterior walls and door. Other boundary conditions including heat source, inlet and outlet surfaces are listed in Table 2.

### 2.2. Modelling of occupant surrounding temperature

The human body shape does influence the indoor thermal environment, and ideally, a human body model should be included in the three-dimensional Multiphysics simulation with updated position changes. However, the objective of this work is to develop an implementable control strategy using limited sensors, making simplification necessary. Therefore, occupants are modelled as vertical line domains in the simulation, defined by the coordinates the lowest and highest points,  $(x, y, z_1)$  and  $(x, y, z_2)$ , respectively. The average temperature along this line represents the temperature around the occupant. This simplification does not compromise the control objective. At each simulation moment, the average temperature on the line domain is calculated as follows:

$$T_{\text{avg}}(t) = \frac{\int_{z_1}^{z_2} T(l) dl}{L} \quad (5)$$

where,  $T_{\text{avg}}(t)$  is the average temperature ( $^{\circ}\text{C}$ ) of the line domain at time  $t$ ;  $T(l)$  is the temperature ( $^{\circ}\text{C}$ ) along the line  $l$  at moment  $t$ ;  $L$  is the total length ( $\text{m}$ ) of the line domain.

To evaluate the effectiveness of using the line model to estimate air temperature around occupants in Multiphysics simulations, this study introduced a cylinder human model into the three-dimensional Multiphysics simulation. The air temperatures obtained from both models were compared across various positions and cases, with detailed settings and results provided in Appendix A. While some differences were observed between the temperatures from the line and cylinder models at certain times, the overall difference remained within an acceptable range. Therefore, the line model will be used in subsequent control system simulations to provide real-time air temperature around the occupants.

### 2.3. Energy consumption calculation

During the operation of the storage electric heater, a portion of the heat generated is stored in the energy storage module. The difference between the electrical energy consumed and the heat energy stored constitutes the energy available for space heating. The energy consumption for space heating can be calculated using Eq. (6).

$$E_{\text{sp}}(t) = \int_{t=0}^{t=t} [P_{\text{HM1}}(t) + P_{\text{HM2}}(t) + P_{\text{HM3}}(t)] dt - E_{\text{st}}(t) \quad (6)$$

where,  $E_{\text{sp}}(t)$  represents the energy consumption ( $\text{kWh}$ ) for space heating from the initial moment up to moment  $t$ ;  $P_{\text{HM1}}(t)$ ,  $P_{\text{HM2}}(t)$ ,  $P_{\text{HM3}}(t)$  are the operating power ( $\text{kW}$ ) of the three heating modules in the storage electric heater at moment  $t$ , respectively;  $E_{\text{st}}(t)$  is the stored energy ( $\text{kWh}$ ) of the electric heater from the initial moment to the moment  $t$ .

The stored energy can be expressed as a function of the amount of

**Table 2**  
Boundary conditions for the Multiphysics simulation model.

| Boundary type | Description      | Condition            | Value                                         |
|---------------|------------------|----------------------|-----------------------------------------------|
| Wall          | Exterior         | Convective heat flux | $2.5 \text{ W}/(\text{m}^2 \bullet \text{K})$ |
| Wall          | Interior         | No slip              | /                                             |
| Door          | Exterior         | Convective heat flux | $3 \text{ W}/(\text{m}^2 \bullet \text{K})$   |
| Heat source   | Heating module 1 | Heat rate            | $P_{\text{HM1}}$                              |
| Heat source   | Heating module 2 | Heat rate            | $P_{\text{HM2}}$                              |
| Heat source   | Heating module 3 | Heat rate            | $P_{\text{HM3}}$                              |
| Inlet         | Fan exhaust      | Velocity             | $V$                                           |
| Outlet        | /                | Pressure             | Static                                        |

temperature change of the storage modules. The material of the storage module in the storage electric heater used in this study is magnetite, which is the sensible heat storage material. The relationship between the stored energy and the amount of temperature change of the storage material can be written by the Eq. (7).

$$E_{st}(t) = M \bullet C_p \bullet (T(t) - T(0)) \quad (7)$$

where,  $M$  is the mass of the material (kg);  $\delta$  is temperature capacity conversion coefficient (C);  $C_p$  is the specific heat capacity at constant stress (J/(kg • K));  $T(t)$  and  $T(0)$  are the temperatures (°C) of the storage module at the initial and time  $t$ , respectively.

### 3. Hierarchical state machine control strategy

Hierarchical State Machine (HSM) decomposes the state machine model of an entire system into multiple layers, each layer being responsible for a specific function or module. Compared to conventional control strategies such as setpoint control and PID control, the HSM control strategy offers several advantages, including enhancing modularity, maintainability, and scalability in system design and operation [28]. This approach allows for a more structured and flexible system, facilitating easier updates and improvements while maintaining performance across different operational conditions. In this paper, the HSM is introduced into the control of storage electric heating system, which benefits greatly from HSM's modularity; a three-layer HSM strategy in association with the operating power of the electric heater is proposed, which is illustrated in Fig. 2.

The storage electric heating system used in this paper is Quantum series electric heater QM150RF which is an advanced product made in the UK. This storage electric heater contains three electric heating modules inside, each module has a working power of 1 kW, and uses a fan to extract the stored heat out from the bottom of the heater into the space. The average air velocity from the fan is 1 m/s. As shown in Fig. 2, the storage electric heater has three modes of operating power, namely Mode 1 (1 kW), Mode 2 (2 kW) and Mode 3 (3 kW).

The storage electric heater can either heat the space using only stored heat or by combining electric heating with stored heat. While using only stored heat conserves electricity, it results in a longer response time to

achieve occupants' thermal comfort. This study considers both energy consumption and response time, establishing two operational states for each mode: fast heating (FH) and normal heating (NH). The switch between these states is driven by temperature feedback: the heater operates in fast heating mode when the temperature change exceeds the set deviation range ( $\Delta T$ ), and in normal heating mode when it remains within the range.

The co-simulation platform to accommodate the HSM control system, the Multiphysics model and the interface for data communication is illustrated in Fig. 3. The hierarchical state machine control strategy receives the feedback temperature and changes at each control time step. Then, it formulates the working power of each heating module in the storage electric heating system and the heat release rate of the heating system. Through real-time interaction, after receiving the actions formulated by the hierarchical state machine control strategy, the Multiphysics model performs a Multiphysics simulation to predict the temperature distribution in the room. After the occupant's position is obtained, the Multiphysics simulation results at that position are extracted and fed back to the control strategy.

The models for evaluating human thermal comfort consider different factors have been widely studied in recent years [29–32]. Among these methods, the PMV-PPD (Predicted Mean Vote-Predicted Percentage Dissatisfied) model is a comprehensive and scientifically validated tool for assessing thermal comfort [33]. It takes a wide range of factors to human perception of thermal environments into account, providing valuable insights for creating comfortable indoor spaces. The detailed calculation process of the PMV-PPD model is provided in Appendix B. This model comprehensively considers the effects of clothing insulation ( $I_{cl}$ ), metabolic rate ( $M$ ), air temperature ( $t_a$ ), mean radiant temperature ( $t_{mrt}$ ), air velocity ( $v$ ), and water vapor pressure ( $P_a$ ) on human thermal comfort, as shown in Eq. (8).

$$[PMV, PPD] = f(I_{cl}, M, t_a, t_{mrt}, v, P_a) \quad (8)$$

When the PMV value is 0, the PPD value is 5 %, indicating a neutral thermal sensation and optimal thermal comfort. This paper assumes that all parameters except for air temperature are known in advance, and then calculates the air temperature corresponding to a neutral thermal sensation based on the PMV-PPD model, as shown in Fig. 4.

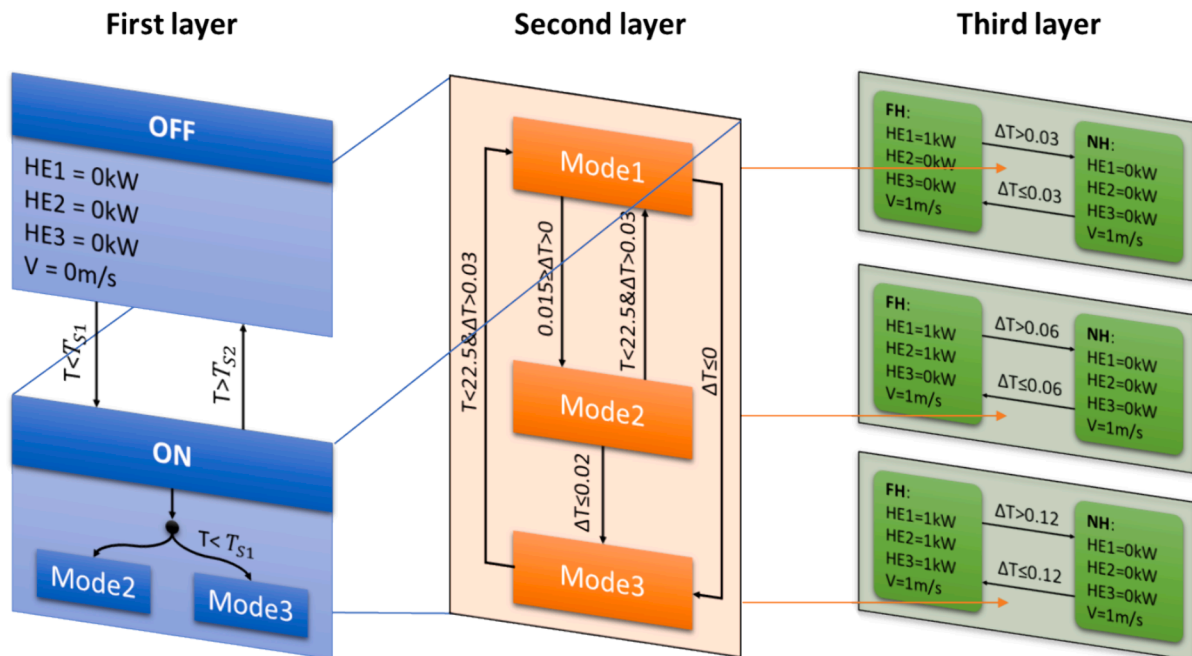


Fig. 2. Hierarchical state machine control strategy.



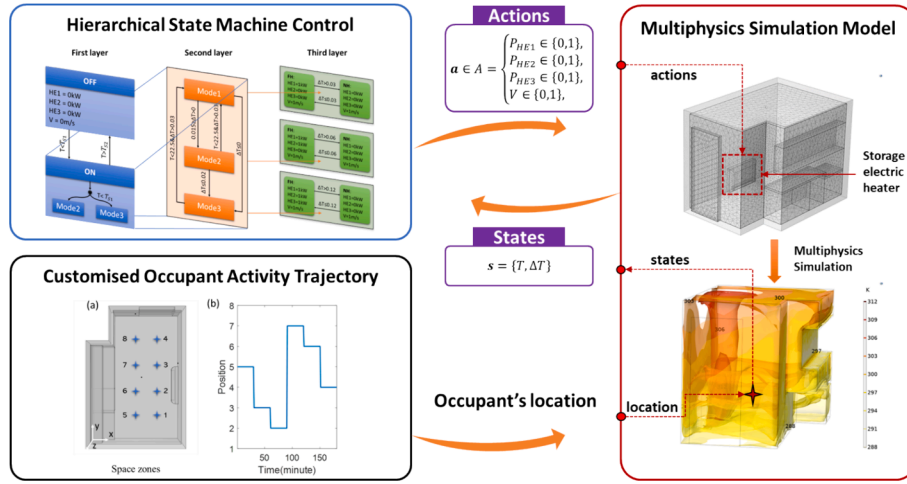


Fig. 3. Illustration of the Multiphysics simulation model derived hierarchical state machine control system.

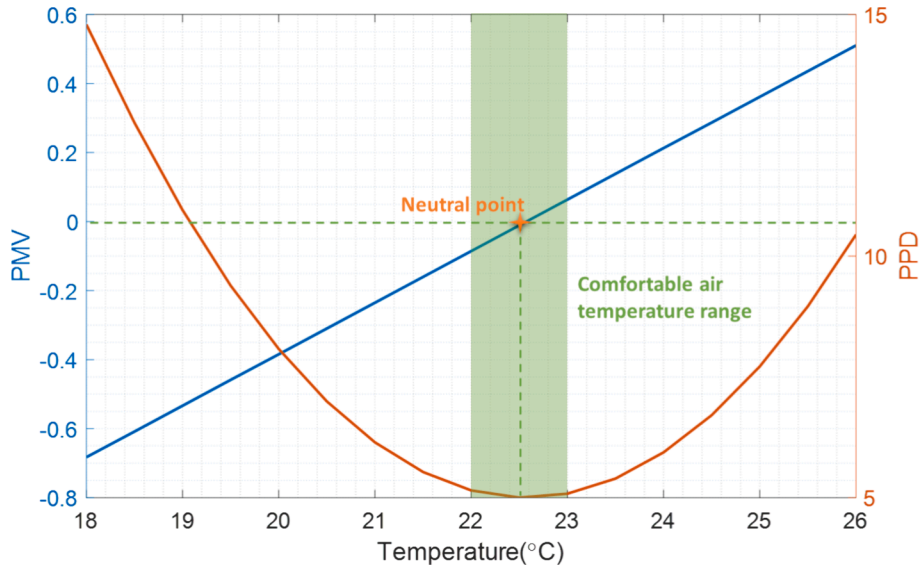


Fig. 4. PMV-PPD model.

To prevent the thermostat from continuously cycling the heater and to minimize heater state transitions, this paper uses the temperature range corresponding to a neutral thermal sensation as the control target, as shown in Fig. 4. Therefore, the HSM aims to maintain the temperature between  $T_{S1}$  and  $T_{S2}$ . In this paper, a case with  $T_{S1}$  and  $T_{S2}$  set to be 22°C and 23°C, respectively, is studied.

This paper develops a hierarchical state machine control strategy aimed at reducing energy consumption, speeding up the response time for occupants' thermal comfort, and maintaining the air temperature within a comfortable range. As a rule-based strategy, the control objectives do not directly influence the real-time control process but guide the setting of state transition thresholds. These thresholds are manually set based on initial simulation results and may not be optimal.

The mathematical definition of the three layers state machine is expressed by set of states, transition conditions and control actions. The set of states can be represented as:

$$S = \begin{Bmatrix} S_1 \\ S_2 \\ S_3 \end{Bmatrix} = \begin{Bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} & S_{23} \\ S_{31} & S_{32} & S_{33} & S_{34} & S_{35} & S_{36} \end{Bmatrix} \quad (9)$$

where,  $S_1$  is the state set of the layer 1 state machine;  $S_2$  is the state set of

the layer 2 state machine.  $S_3$  is the state set of the layer 3 state machine;  $S_{11}$  and  $S_{12}$  denote the ON and OFF states of the first layer state machine, respectively;  $S_{21}$ ,  $S_{22}$ , and  $S_{23}$  denote the second layer state machine Mode 1, Mode 2, and Mode 3 states, respectively;  $S_{31}$ ,  $S_{32}$ ,  $S_{33}$ ,  $S_{34}$ ,  $S_{35}$ , and  $S_{36}$  denote the FH and NH states of the third layer state machine in different modes of operation, respectively. The transition conditions for moving from one state to another are:

$$T = \begin{cases} T_{mn} : S_m \rightarrow S_n, m, n, i \in \{1, 2, 3\}; j, k \in \{1, 2, 3, 4, 5, 6\} \\ T_{ijk} : S_{ij} \rightarrow S_{ik} \end{cases} \quad (10)$$

where,  $T_{mn}$  is the transfer condition of the  $m$  layer state machine  $S_m$  to the  $n$  layer state machine  $S_n$ ;  $T_{ijk}$  is the transition condition from the  $J_{th}$  state  $S_{ij}$  of the layer  $i$  state machine to the  $K_{th}$  state  $S_{ik}$  of the layer  $i$  state machine. The corresponding control actions are described below:

$$A = [PHE1 \in [0, 1], PHE2 \in [0, 1], V \in [0, 1]] \quad (11)$$

where,  $P_{HE1}$ ,  $P_{HE2}$  and  $P_{HE3}$  are the working power of the three electric heating modules respectively;  $V$  is the heat release rate of the storage electric heater; Power and speed are binary variables. Power takes values in the numbers 0 kW and 1 kW. Rate takes values before 0 m/s and 1 m/s.

With the proposed control strategy, the results are compared with those achieved by the fixed temperature sensor control strategy. The temperature sensors are fixed at the doorway (HSMC@d) and near to the heater (HSMC@h). The occupant-centred space heating control system proposed in this paper is abbreviated to HSMC@o.

Occupant thermal comfort is the primary control objective, and the comfort interval set in this paper is 22°C to 23°C. In the simulation process, the occupants are considered uncomfortable when the temperature around them is outside this interval. To compare the thermal comfort levels of the occupants with different control systems, this paper establishes a thermal discomfort equation as shown in Eq. (12). The thermal discomfort equation calculates the proportion of total hours of use that the occupant experiences thermal discomfort during the use of the electric heating system.

$$\ell(t) = \frac{\int_0^t \text{time}(t)|_{\text{outofthecomfortzone}} dt}{\int_0^t \text{time}(t) dt} \times 100\% \quad (12)$$

where,  $\ell(t)$  represents the proportion of thermal discomfort at time  $t$ .

#### 4. Experimental system setup

An experimental test system is established including NI CompactRIO controller, NI thermocouple input module, NI voltage output module, sensors, driving circuit, and storage electric heater to validate the numerical accuracy of built Multiphysics simulation model. The structure of the test system is shown in Fig. 5, where K-type thermocouple sensors are deployed in the test room and the temperature is measured through NI 9213 thermocouple input module. The power supply for the heater's three heating modules and fan is connected to the NI 9263 voltage analog output module through a drive circuit which consists of four operational amplifiers, four solid state relays and a power supply. The thermocouple input module and voltage analog output module are connected to the NI CompactRIO controller. Finally, the temperature data collection and heater control programs are coded in LabVIEW and a user interface (UI) front panel is designed to interact with the components.

During the test, the sensor temperature data is monitored and collected through the LabVIEW control UI by switching on the fan and different heating modules of the heater respectively. Temperature data of the indoor space were collected under different heater modes and mode transitions. A total of six tests were conducted, each lasting for one hour, with several days between each test. The six tests were as: (1) operate the heater at 1 kW power; (2) operate the heater at 2 kW power; (3) operate the heater at 3 kW power; (4) operate the heater at 3 kW power, then switch to 2 kW power; (5) operate the heater at 3 kW power, then switch to 1 kW power; (6) operate the heater at 2 kW power, then switch to 1 kW power. Finally, the collected temperature data from the

experiments were compared with the corresponding Multiphysics simulation results to verify the numerical accuracy of the simulation model.

#### 5. Experiment test, simulation study and results analysis

This research is simulated based on multi-software platform co-simulation. The process of multi-software platform co-simulation is shown in Fig. 6. First, the 3D model is built in INVENTOR software, which fed the dimension information to COMSOL via the LiveLink interface. Then multiple physical fields and boundary conditions are added to the 3D model in COMSOL to simulate the temperature change in space during the operation of the electric heating system. Then, the hierarchical state machine control strategy is built in SIMULINK software, and LiveLink interface realizes the real-time interaction between SIMULINK and the Multiphysics simulation model in COMSOL. Finally, the co-simulation is completed in SIMULINK software environment. During the co-simulation phase, the simulation interval was set to one minute.

All simulations for this study were conducted on computers equipped with AMD Ryzen 5600H 3.30 GHz processors, 16 GB of RAM, NVIDIA GeForce GTX 1650 GPU, and 64-bit Windows operating systems. Three control systems are studied, namely HSMC@h, HSMC@d, and HSMC@o. In the Multiphysics simulation model, the indoor space is divided into eight regions, with the centre positions of each zone shown in Fig. 7(a). During the experimental validation, thermocouple temperature sensors were fixed at these centre positions. The coordinates of the eight sensors in this space are (1910, 810, 1610), (1910, 1510, 1610), (1910, 2210, 1610), (1910, 2910, 1610), (1310, 810, 1610), (1310, 1510, 1610), (1310, 2210, 1610), and (1310, 2910, 1610). In order to test the performance of different space heating systems, it is necessary to provide a trajectory of the occupants' activities in the indoor space. In this paper, the activity trajectory of a resident is obtained by random ordering: 5->3->2->7->6->4. Assuming that the occupants stay in each area for half an hour, the resulting time-location path is shown in Fig. 7(b). The trail takes a total of 3 h.

##### 5.1. Mesh independent validation

A mesh independence study is performed to find the optimal mesh size and minimize the computational cost without compromising the solution accuracy. Physics controlled meshing is implemented to generate the mesh. Seven physics controlled meshes namely fine, normal, coarse, coarser, extra coarse, and extremely coarse are generated and compared. For the comparison, the fine mesh is set as a benchmark and other meshes are compared with it. The error at eight central spatial locations shown in Fig. 7(a) is calculated for the other

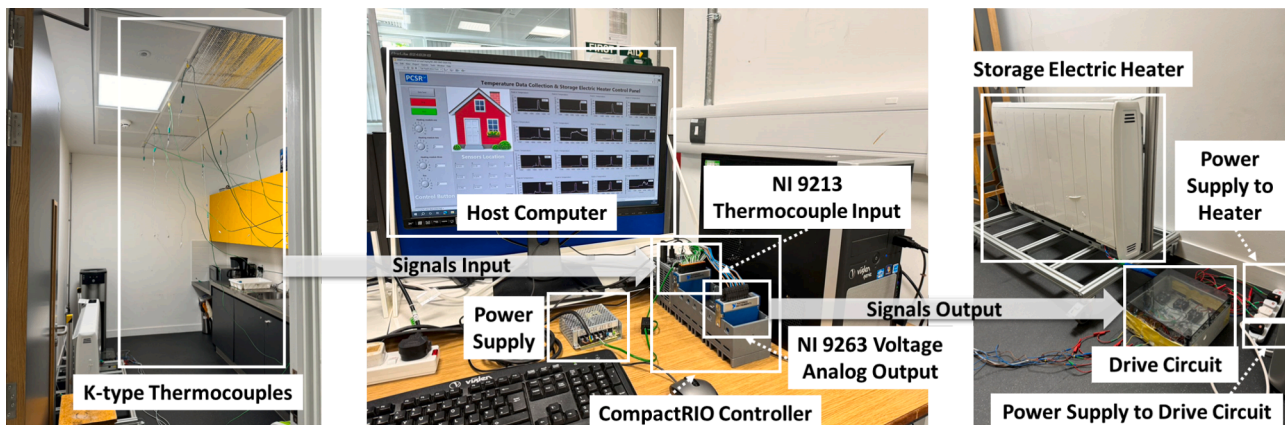


Fig. 5. Schematic diagram of the experimental setup.

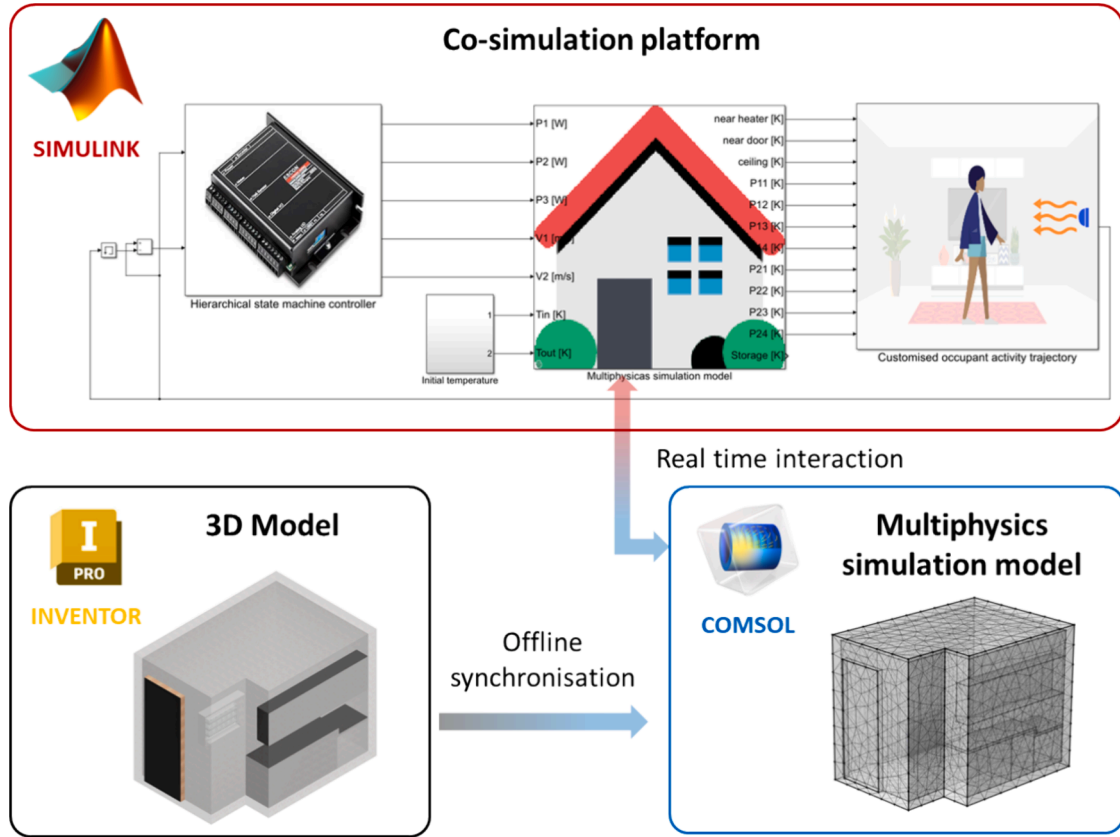


Fig. 6. Demonstration of multi-software co-simulation.

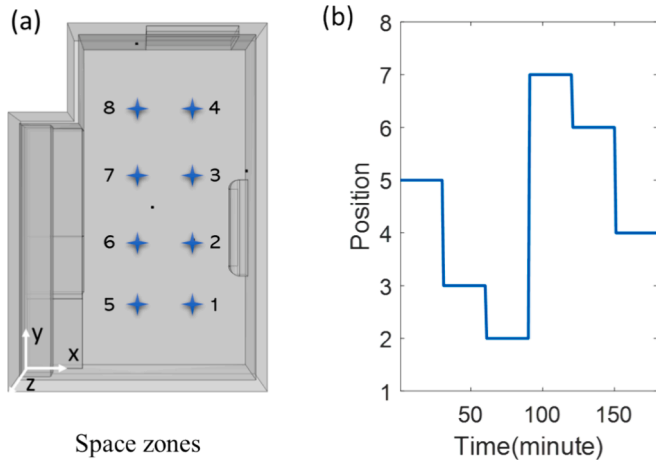


Fig. 7. Occupant activity trajectory: (a) Space zones; (b) Customized trajectory.

mesh sizes with respect to fine mesh, as shown in Eq. (13).

$$\mathcal{L}_{err}(t)|_{P_i}^{mesh} = \frac{|T(t)|_{P_i}^{mesh} - T(t)|_{P_i}^{Fine}}{T(t)|_{P_i}^{Fine}} \times 100\% \quad (13)$$

where,  $mesh \in \{Normal, Coarse, Coarser, Extra\ coarse, Extremely\ coarse\}$ ;  $P_i$  is the position of the probe,  $i \in \{1, 2, 3, \dots, 8\}$ ;  $\mathcal{L}_{err}(t)|_{P_i}^{mesh}$  is the temperature error at the  $P_i$  position for the comparison grid at time  $t$ ;  $T(t)|_{P_i}^{mesh}$  and  $T(t)|_{P_i}^{Fine}$  represent the temperature ( $^{\circ}\text{C}$ ) at the  $P_i$  position at time  $t$  for the comparison mesh and the benchmark mesh, respectively.

To save computation time, the simulation horizon of each case in the grid independence study is 15 –minute. The simulation step size is set to

1 min. Based on Eq. (13), the simulation errors at eight space locations under different grid densities are shown in Fig. 8.

The number of elements and the time required to complete the scenario simulations for different grid settings are listed in Table 3. The average errors at the eight locations when the physical field controlled the grid settings NORMAL, COARSE, COARSER, EXTRA COARSE, and EXTREMELY COARSE were 3.693 %, 3.732 %, 3.751 %, 9.926 %, and 9.064 %, respectively. Taking into consideration both simulation accuracy and computational efficiency, this study uses coarser setting to ensure the simulation speed without jeopardizing simulation accuracy.

## 5.2. Numerical validation

Based on the experimental test rig, six tests were completed in this paper to collect temperature data at 11 locations in the room. The 11 spatial locations are: sensor 1 is arranged on the ceiling; sensor 2 is arranged to the doorway; sensor 3 is arranged near the heater; and sensors 4–11 are arranged at each of the 8 locations specified in Fig. 7 (a). The duration of temperature data collection is one hour. Validation of the numerical accuracy of the Multiphysics simulation model is accomplished through error analysis of the experimental and simulation data. During the heater operation, the numerical error between the experimental data and the simulation data is calculated by Eq. (14), and the results are shown in Fig. 9. The numerical errors observed across various cases are summarized in Table 4. For all heater operation modes, the maximum numerical error at the 11 sensor positions was below 5 %, with an average numerical error of under 1 %. These results indicate that the Multiphysics simulation model developed in this paper accurately simulates indoor temperature distribution changes.

$$\mathcal{L}_{Nerr}(t)|_{sp(k)} = \frac{|T(t)|_{sp(k)}^{experiment} - T(t)|_{sp(k)}^{simulation}}{T(t)|_{sp(k)}^{experiment}} \times 100\% \quad (14)$$

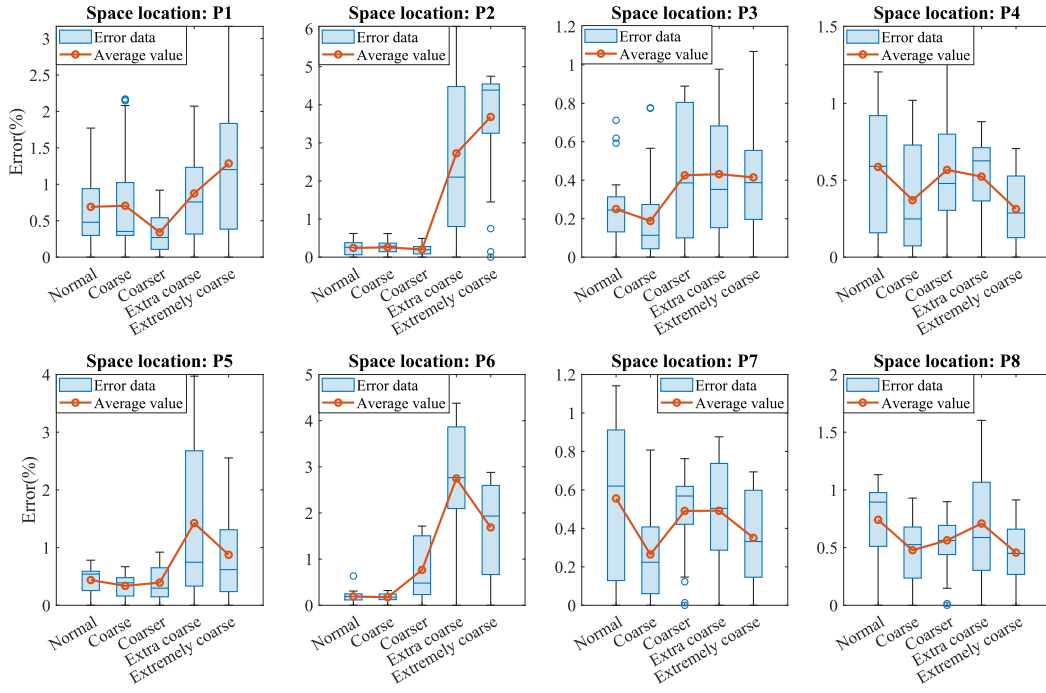


Fig. 8. Mesh independent validation of the Multiphysics simulation model.

Table 3

Grid independence verification (15-minute scenario simulation).

| Mesh settings    | Number of elements | Computation time (s) | Average error (%) |
|------------------|--------------------|----------------------|-------------------|
| Fine (benchmark) | 627,042            | 13,840               | /                 |
| Normal           | 225,639            | 3860                 | 3.693             |
| Coarse           | 103,545            | 690                  | 3.732             |
| Coarser          | 38,662             | 199                  | 3.751             |
| Extra coarse     | 19,033             | 76                   | 9.926             |
| Extremely coarse | 12,098             | 49                   | 9.064             |

where,  $k$  is the position of the sensor,  $k \in \{1, 2, 3, \dots, 11\}$ ;  $\mathcal{L}_{Nerr}(t)|_{sp(k)}$  is the numerical accuracy at the  $k$ th sensor position at time  $t$ ;  $T(t)|_{sp(k)}^{experiment}$  is the experimental data ( $^{\circ}\text{C}$ ) at the  $k$ th sensor position at time  $t$ ;  $T(t)|_{sp(k)}^{simulation}$  is the simulation data ( $^{\circ}\text{C}$ ) at the  $k$ th sensor position at time  $t$ .

Mean square error (MSE) error shown as Eq. (15) was used to quantify the numerical accuracy at the different locations of the sensors. The MSE across various cases are summarized in Table 4. Across all power levels, the maximum and average MSE at the 11 sensor locations were below  $0.21^{\circ}\text{C}$  and  $0.06^{\circ}\text{C}$ , respectively. These results confirm that the Multiphysics simulation model established in this study can effectively simulate the indoor temperature changes during the heater's operation.

$$\mathcal{L}_{MSE}|_{sp(k)} = \frac{1}{N} \sum_{t=1}^N \left( T(t)|_{sp(k)}^{experiment} - T(t)|_{sp(k)}^{simulation} \right)^2 \quad (15)$$

where,  $\mathcal{L}_{MSE}|_{sp(k)}$  is the MSE at the  $k$ th sensor position ( $^{\circ}\text{C}$ ).

During the operation of the control system, the Multiphysics simulation model provides real-time temperature information near the occupants to the control strategy. Simulation errors can lead to inappropriate control signals, potentially affecting the occupants' actual thermal comfort. While simulation errors are inevitable, the accuracy of the model established in this paper has been verified to be within an

acceptable range. Although these errors may occasionally affect thermal comfort, the simulation model generally provides reliable temperature information to the control strategy.

### 5.3. Analysis of thermal comfort results

The results of thermal comfort for occupants under different control systems are presented in Fig. 10. Fig. 10 (a), (d), and (f) depict temperature data extracted from the fixed-position sensors and temperature data at locations of occupant activity trajectories. The objective of the control strategy is to maintain the temperature surrounding the occupant within the comfort range of  $22^{\circ}\text{C}$ – $23^{\circ}\text{C}$ . During the three-hour simulation, temperatures at occupant activity locations under the HSMC@h control system frequently fell below the comfort range. Under the HSMC@d control process, temperatures at occupant activity locations were often above the comfort range. In the HSMC@o control process, temperatures at occupant activity locations were generally maintained within the comfort range. The thermal discomfort of occupants during the control process was quantified using a thermal discomfort formula, as shown in Fig. 10(e). Due to the initial temperature in the room being below the comfort range, the electric heating system required some response time to elevate the temperature near the occupants to the thermal comfort zone. Therefore, during the early stages of the control system operation, the thermal discomfort of occupants was 100 %, which gradually decreased with increasing operating time for all three control systems. After 40 min, the thermal discomfort of occupants exhibited different trends. In the subsequent control process of HSMC@h, the thermal discomfort of occupants gradually increased. In the subsequent control process of HSMC@d, the thermal discomfort of occupants showed significant fluctuations. In the subsequent control process of HSMC@o, the thermal discomfort of occupants gradually decreased. At the end of the 3-hour simulation, the thermal discomfort indices for occupants under HSMC@h, HSMC@d, and HSMC@o were 57.18 %, 37.78 %, and 19.44 %, respectively. Compared to HSMC@h and HSMC@d, the thermal discomfort of occupants decreased by 37.74 % and 18.34 %, respectively. Fig. 10(b) shows that the response time for the temperature near occupants to rise from the initial temperature to the comfort zone was similar under the three



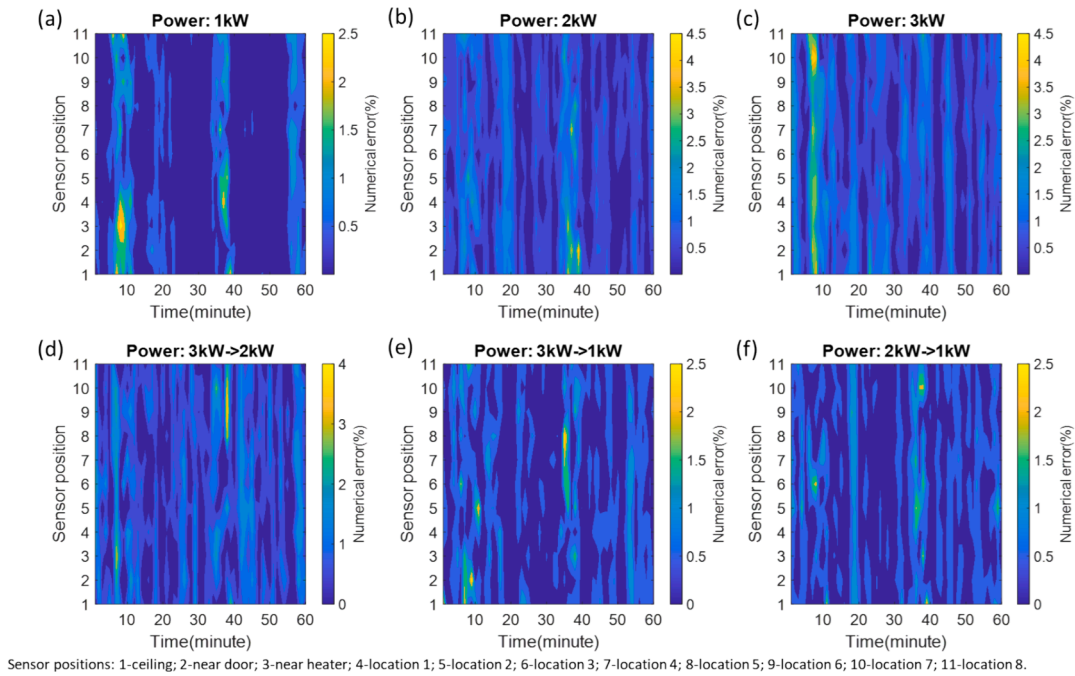


Fig. 9. Numerical validation of the Multiphysics simulation model.

**Table 4**  
Error values across different cases.

| Error Cases                         | Maximum numerical error (%) | Average numerical error (%) | Maximum MSE (°C) | Average MSE (°C) |
|-------------------------------------|-----------------------------|-----------------------------|------------------|------------------|
| 1 kW (one hour)                     | 2.69                        | 0.44                        | 0.0793           | 0.0190           |
| 2 kW (one hour)                     | 4.60                        | 0.83                        | 0.1834           | 0.0520           |
| 3 kW (one hour)                     | 4.55                        | 0.97                        | 0.2006           | 0.0589           |
| 3 kW (half hour) → 2 kW (half hour) | 4.24                        | 0.82                        | 0.0981           | 0.0461           |
| 3 kW (half hour) → 1 kW (half hour) | 2.87                        | 0.53                        | 0.0494           | 0.0207           |
| 2 kW (half hour) → 1 kW (half hour) | 2.53                        | 0.53                        | 0.0489           | 0.0192           |

control systems. The simulation used a fixed step time discretization with a time interval set to 1 min. Statistics on the number of state transitions during the 3-hour simulation process are presented in Fig. 10 (c), with HSMC@h, HSMC@d, and HSMC@o having 140, 153, and 132 state transitions, respectively. Frequent state transitions can lead to frequent start-stop cycles of the heating module, resulting in additional energy losses. The number of state transitions under HSMC@o decreased by 5.71 % and 13.73 % compared to HSMC@h and HSMC@d, respectively. These simulation results indicate that the proposed HSMC@o can enhance the thermal comfort of occupants during the heating system operation and reduce the number of heating system state transitions.

#### 5.4. Analysis of energy consumption

The operating power of the storage electric heater with different control scheme settings is illustrated in Fig. 11. The simulations in this

study were conducted based on a fixed simulation step of 1 min. Fig. 11 (a) depicts the variation of the heater's operating power over time, while Fig. 11(b) provides a statistical analysis of the duration of the electric heater operation at different power levels over the three-hour simulation period. In the HSMC@h control process, the electric heater tends to operate at lower power levels, with a total duration of 26 min at 1 kW, which is 2.89 and 1.86 times longer than that under HSMC@d and HSMC@o, respectively. In the HSMC@d control process, the electric heater tends to operate at higher power levels, with a total duration of 71 min at 3 kW, which is 2.96 and 1.48 times longer than that under HSMC@h and HSMC@o, respectively. In the HSMC@o control process, the electric heater operates at different power levels with a relatively balanced time distribution.

The energy consumption of the three control systems is illustrated in Fig. 12. A portion of the electrical energy consumed by the storage electric heater is allocated for space heating, while the remaining part is stored within the internal storage module. Fig. 12 (a), (b), and (c) display the electrical energy consumed and the thermal energy stored during the operational processes of the three control strategies. The difference between the consumed electrical energy and stored thermal energy represents the energy utilized for space heating. After 3 h of operation, the energy consumption for HSMC@h, HSMC@d, and HSMC@o is 1.0306 kWh, 2.1197 kWh, and 1.6115 kWh, respectively. HSMC@h exhibits the most favorable energy consumption performance among these three control systems. The energy consumption of HSMC@o is 1.56 times that of HSMC@h and 76 % of that of HSMC@d.

Based on a comprehensive analysis of simulation results, the HSMC@o control system provides superior thermal comfort for occupants and suboptimal operational economy for space heating. While the HSMC@h control system demonstrates excellent operational economy in space heating, it exhibits the highest thermal discomfort index. The HSMC@d control system has the poorest operational economy for space heating, and the performance of HSMC@o is the most favorable.

#### 6. Concluding remarks

This paper proposed a new thermal comfort driven space heating control via hierarchical state machine (HSM) control strategy interacting with Multiphysics simulation. The control system is implemented via

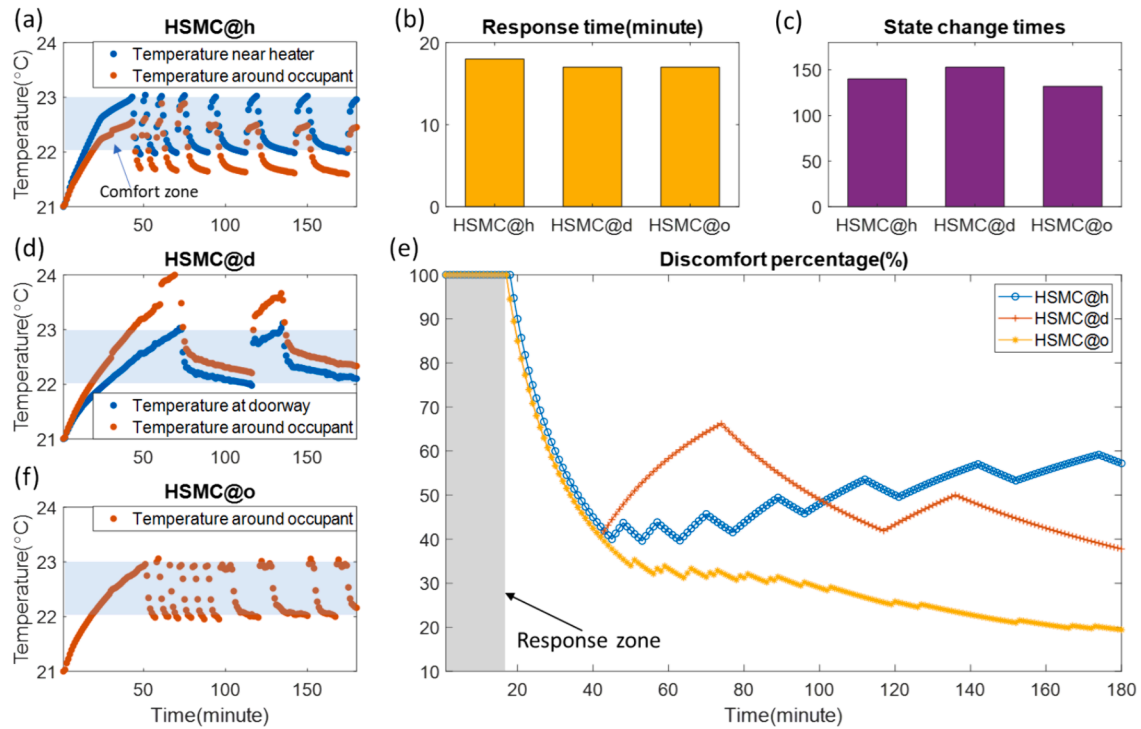


Fig. 10. Simulation results of occupant thermal comfort under HSMC@h, HSMC@d and HSMC@o.

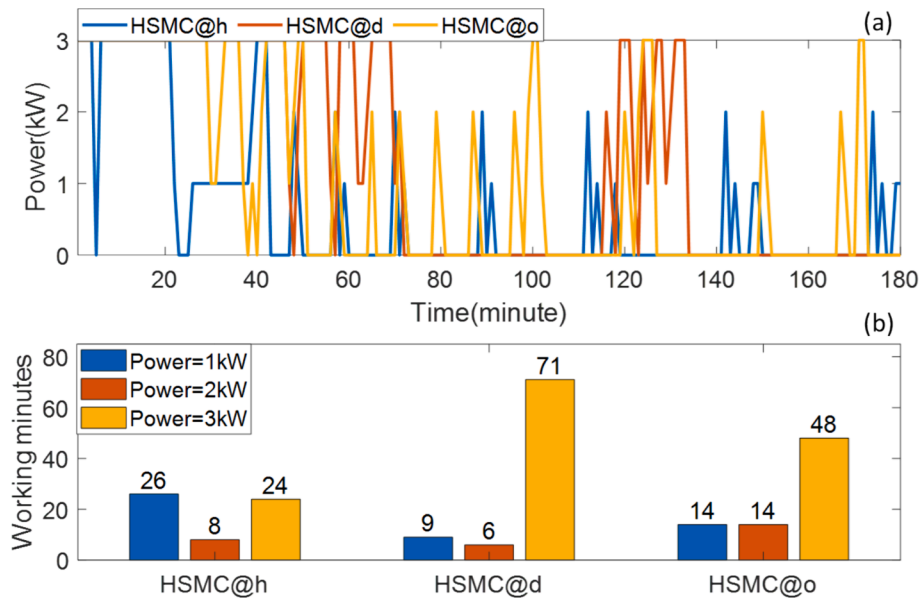


Fig. 11. Operating power of storage electric heater under HSMC@h, HSMC@d and HSMC@o.

a co-simulation platform which accommodates the software from the space dimension INVENTOR model to COMSOL thermodynamics model. Then the platform integrates the models with the SIMULINK which is used for developing the HSM control strategy. From the study, Multiphysics 3D simulation modelling reveals the inhomogeneous nature of temperature distribution in a bounded space, which provides the sounding evidence to support the occupant thermal experience control philosophy. An experimental test rig was set up to verify the numerical accuracy of the Multiphysics simulation model.

The experimental results show that the maximum and average MSE error of the test data and simulation data is less than 0.21°C and 0.06°C at 11 locations in the room under different operating modes of the

heater. It confirms that the Multiphysics simulation model built in this paper has a good accuracy in simulating the change of indoor temperature distribution. The case studies were analysed in this paper, and the results demonstrates a clear energy saving, potential over 20 %, by using the space heating control strategy proposed in this paper. Meanwhile, by analysing the extracted temperature data following the occupant's position trajectory, the HSM control strategy proposed in this paper reduces the thermal discomfort of the occupants by up to 37.74 %. Real-time interaction of control strategies and the Multiphysics simulation model can be achieved through a co-simulation platform in this paper, which provides a methodology for the development of new space heating control systems.

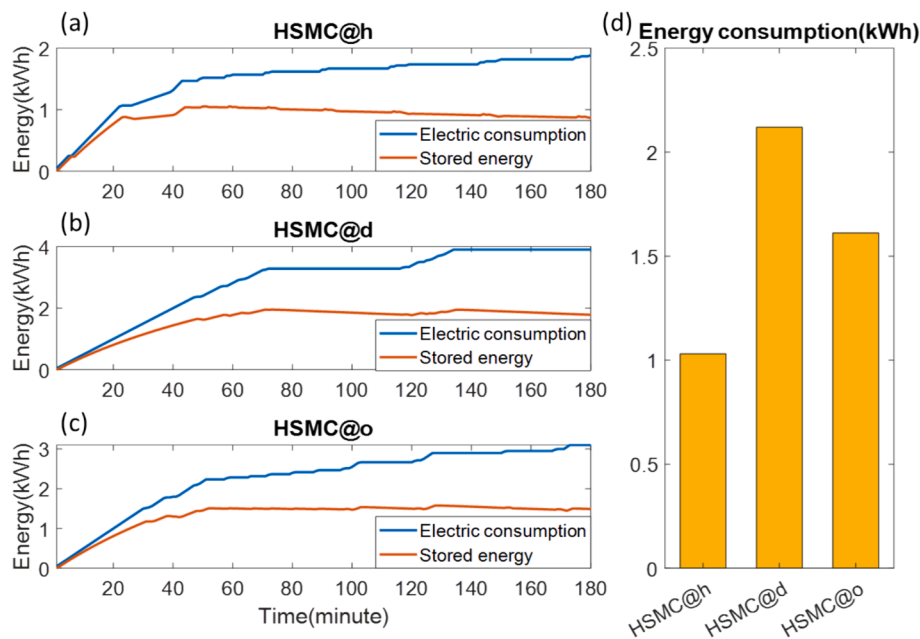


Fig. 12. Energy consumption for space heating under HSMC@h, HSMC@d and HSMC@o.

This study provides several insights into space heating control methods but has some limitations. First, the Multiphysics simulation model was developed using COMSOL, a commercial software that requires significant computational power and is time-consuming to solve. Additionally, the simulations and tests were conducted in a prototype kitchen within a laboratory setting, with no validation in other room types. Future research will address these limitations to improve the robustness and applicability of the proposed method.

#### CRedit authorship contribution statement

**Yunfei Bai:** Writing – original draft, Visualization, Methodology, Formal analysis, Data curation. **Chenghao Li:** Writing – review & editing, Software. **Sunku Prasad Jenne:** Writing – review & editing, Software. **Sizhe Zhang:** Software. **Jihong Wang:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

#### Appendix

The appendix is divided into two sections. Section A compares the results of the physics simulation using both the human body model and the line model. Although the line model shows some errors at certain times, its use significantly improves the operational efficiency of the control system. Section B details the PMV-PPD thermal comfort evaluation model. In this model, once other parameters are assigned values, a range of air temperatures can be determined based on the thermal comfort index, which then serves as the control target.

##### A. Comparison of results between cylinder and line models of the human body in physics simulation

A Cylinder is one of the simplest ways to represent the human body and it is often used in thermal field simulations [34–36]. However, it is typically used in physical field simulations where the position of the human body is fixed. The position of the human body is in reality moving frequently, and to obtain the temperature around the body during this process, the Cylinder human body model needs to change with the position of the body in the physical field simulation. However, in continuous Multiphysics simulations, a change in the position of the Cylinder model will inevitably lead to a change in the solution mesh, and the position of the Cylinder model cannot be set as a controlled variable in COMSOL because it is a parameter that constitutes the geometry. Although moving mesh can contribute to the solution of the problem, its introduction can complicate the solution and greatly increase the solution time. In reality, the temperature around the human body is measured using temperature sensors, but deploying a large number of sensors in a space is not feasible. Therefore, this paper explores adding line domains in a 3D spatial Multiphysics simulation model and using its average temperature to approximate the air temperature around the human body. Since the line model does not account for the human body's impact on the thermal environment, this paper conducts Multiphysics simulations with both the line and cylinder human body model, comparing the results to analyse the performance of the line model in simulations.

According to reference [36], the cylinder human body model has a height of 1.6 m, a radius of 0.14 m, a surface area of 1.41 m<sup>2</sup> and is suspended 10 cm above the ground. In order to obtain the air temperature around the human body, this paper adds a cylinder surface 1 cm outside the human body model, as shown in Fig. A1(a). This approach ignores the different thermal sensitivities of various body parts and uses the average temperature of the cylinder surface as the air temperature around the human body, as indicated in Eq. (A1).

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

#### Data availability

Data will be made available on request.

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$$\bar{T}_{cy} = \frac{1}{2\pi h} \int_0^h \int_0^{2\pi} T(R\cos\theta, R\sin\theta, z) d\theta dz \quad (A1)$$

where,  $\bar{T}_{cy}$  represents the average temperature (C) of the air cylinder surface;  $R$  and  $h$  are the radius (m) and height (m) of the air cylinder, respectively;  $T(R\cos\theta, R\sin\theta, z)$  denotes the temperature (C) distribution on the air cylinder surface.

Furthermore, this section aims to investigate the impact of human body geometry on the simulation. To simplify the calculation, the cylinder human body model is set to be adiabatic. The visualization of the cylinder human body model in 3D space is shown in Fig. A1(b).

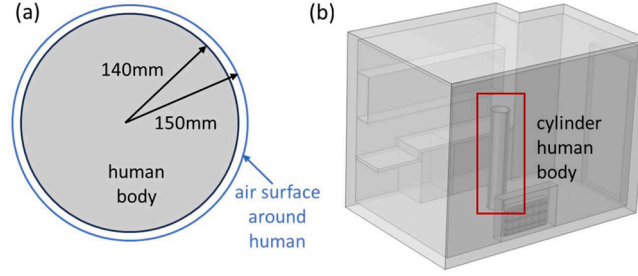


Fig. A1. Cylinder human body model: (a) top view (schematic); (b) visualization in 3D space.

Change the position of the cylinder model in space, and at each position, simulate the temperature variations when the electric heater is operated at 1 kW, 2 kW, and 3 kW, respectively. Then, compare the temperature variations of the line model and the cylinder model at the same positions. The comparison results are shown in Figs. A2, A3, and A4. Numerical error and mean squared error between the two models were calculated using Eq. (14) and (15). During the operation of the electric heater at 1 kW, 2 kW, and 3 kW, the maximum numerical errors between the line model and the cylinder model at different positions were 1.1 %, 1.4 %, and 1.1 %, respectively. The maximum mean squared errors at different positions were 2.0 %, 2.6 %, and 2.4 %, respectively.

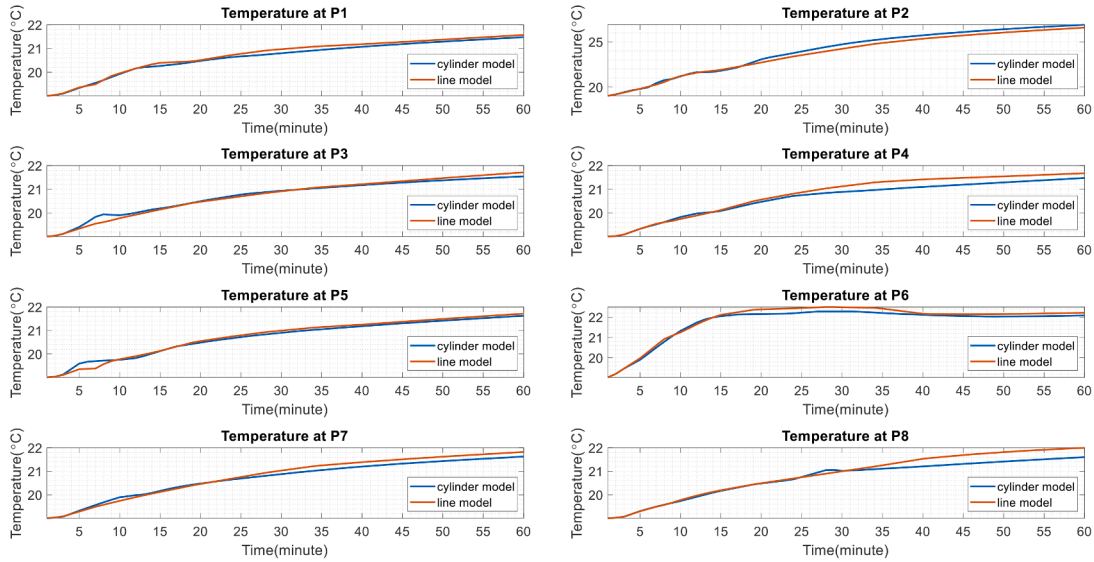


Fig. A2. Comparison of temperature variations of line model and cylinder model at different position when electric heater is operated at 1 kW.



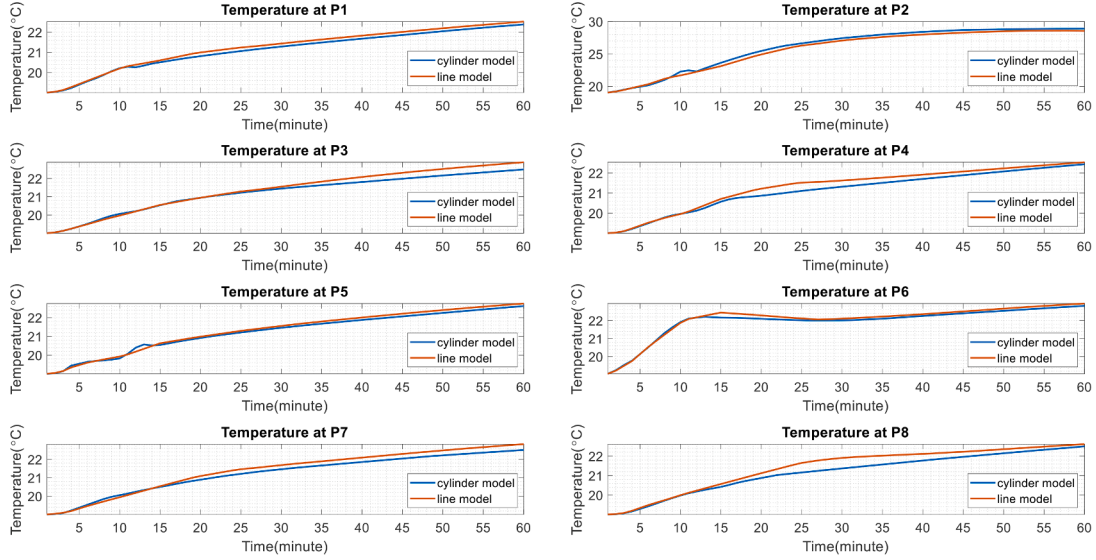


Fig. A3. Comparison of temperature variations of line model and cylinder model at different position when electric heater is operated at 2 kW.

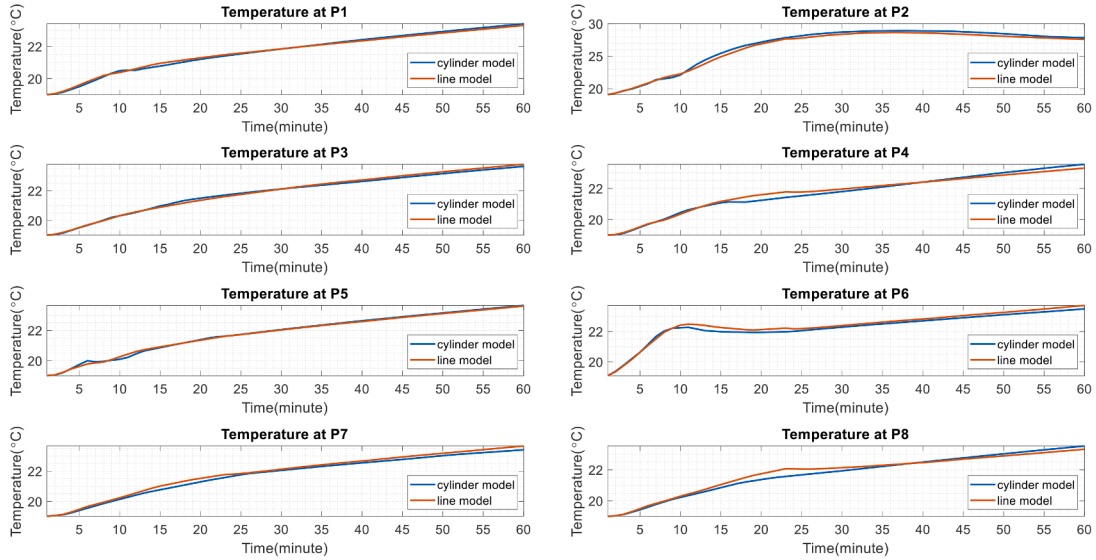


Fig. A4. Comparison of temperature variations of line model and cylinder model at different position when electric heater is operated at 3 kW.

### B. PMV-PPD thermal comfort evaluation method.

The PMV (Predicted Mean Vote)-PPD (Predicted Percentage of Dissatisfied) model, widely used in the field of thermal comfort research, evaluates the comfort level of indoor environments. This model provides a standardized method to predict human thermal comfort based on various environmental and personal factors.

The PMV index evaluates the thermal environment by measuring six variables: two personal variables including clothing insulation ( $I_{cl}$ ) and metabolic rate ( $M$ ), and four environmental parameters including air temperature ( $t_a$ ), mean radiant temperature ( $t_{mrt}$ ), air velocity ( $v$ ), and water vapor pressure ( $P_a$ ). The relationships between these parameters are described by the following equation [37]:

$$PMV = [0.303 \cdot \exp(-0.036 \cdot M) + 0.028] \cdot \left\{ (M - W) - 3.05 \cdot 10^{-3} \cdot [5733 - 6.99 \cdot (M - W) - P_a] - 0.42 \cdot [(M - W) - 58.15] - 1.7 \cdot 10^{-5} \cdot M \right. \\ \left. \cdot (5867 - P_a) - 0.0014 \cdot M \cdot (34 - t_a) - 3.96 \cdot 10^{-8} \cdot f_{cl} \cdot [(t_{cl} + 273)^4 - (t_{mrt} + 273)^4] - f_{cl} \cdot h_c \cdot (t_{cl} - t_a) \right\} \quad (A2)$$

where:

$$t_{cl} = 35.7 - 0.028 \cdot (M - W) - I_{cl} \cdot \left\{ 3.96 \cdot 10^{-8} \cdot f_{cl} \cdot [(t_{cl} + 273)^4 - (t_{mrt} + 273)^4] + f_{cl} \cdot h_c \cdot (t_{cl} - t_a) \right\} \quad (A2a).$$

$$h_c = \begin{cases} 2.38 \cdot |t_{cl} - t_a| & \text{if } 2.38 \cdot |t_{cl} - t_a|^2 \geq 12.1 \cdot \sqrt{v} \\ 12.1 \cdot \sqrt{v} & \text{if } 2.38 \cdot |t_{cl} - t_a|^2 < 12.1 \cdot \sqrt{v} \end{cases} \quad (A2b).$$

$$f_{cl} = \begin{cases} 1.00 + 1.290 \bullet I_{cl} \text{ if } I_{cl} \leq 0.078 m^2 \bullet kWh \\ 1.05 + 0.645 \bullet I_{cl} \text{ if } I_{cl} > 0.078 m^2 \bullet kWh \end{cases} \quad (A2c).$$

where,  $M$  is the metabolic rate, in watts per square metre ( $W/m^2$ );  $W$  is the effective mechanical power, in watts per square metre ( $W/m^2$ );  $I_{cl}$  is the clothing insulation, in square metres kelvin per watt ( $m^2 \bullet K/W$ );  $f_{cl}$  is the clothing surface area factor;  $t_a$  is the air temperature, in degrees Celsius ( $^{\circ}C$ );  $t_{mrt}$  is the mean radiant temperature, in degrees Celsius ( $^{\circ}C$ );  $v$  is the relative air velocity, in metres per second ( $m/s$ );  $P_a$  is the water vapour partial pressure, in pascals ( $Pa$ );  $h_c$  is the convective heat transfer coefficient, in watts per square metre kelvin [ $W/(m^2 \bullet K)$ ];  $t_{cl}$  is the clothing surface temperature, in degrees Celsius ( $^{\circ}C$ ).

The relationship between PMV and thermal sensation is listed in Table A1 [33].

**Table A1**

Relationship between PMV and thermal sensation.

| Thermal sensation | Cold | Cool | Slightly cool | Neutral | Slightly warm | Warm | Hot |
|-------------------|------|------|---------------|---------|---------------|------|-----|
| PMV               | −3   | −2   | −1            | 0       | 1             | 2    | 3   |

The PPD index complements the PMV by predicting the percentage of people likely to feel thermally uncomfortable in a given environment. It is derived from the PMV value and provides a quantitative measure of discomfort. The relationship between PMV and PPD is formulated as follows:

$$PPD = 100 - 95 \bullet \exp(-0.003353 \bullet PMV^4 - 0.2179 \bullet PMV^2) \quad (A3)$$

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